Shreya Katiyar

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Education

University of California, Irvine — Irvine, CA
B.S. Informatics, Minor in Innovation & Entrepreneurship — *Expected June 2026*

Experience

AI Lead – Senior Capstone: MLC Chatbot — January 2026 – Present
Prenostik, UC Irvine — *Irvine, CA*

- Trained and evaluated models on real student health data to identify at-risk patterns; benchmarked LLMs on latency, accuracy, and cost to select optimal model for production.
- Built retrieval pipeline using embeddings and vector search to rank and surface the most relevant student records – directly analogous to retrieval and relevance ranking in ads systems.
- Designed NL-to-SQL layer converting natural language queries into structured database calls; fed schema-aware retrieved context into LLM for grounded, personalized responses.
- Led cross-functional team of 5; delivered live demos communicating model behavior and tradeoffs to both technical and non-technical stakeholders.

AI Research Developer — June 2024 – March 2025
Automated Grading & Feedback Lab, UC Irvine — *Irvine, CA*

- Used embeddings and semantic similarity scoring to rank student responses against rubric criteria – a retrieval and relevance problem applied to educational data at 200+ submission scale.
- Developed RAG architecture vectorizing rubric content into a searchable knowledge base; reduced hallucinations by grounding model outputs in retrieved instructor content.
- Deployed full pipeline on AWS (EC2, S3, Lambda) and Google Cloud; iterated on model behavior based on instructor feedback across UCI and UCSD.

Data Analytics Extern — May 2025 – September 2025
Beats by Dre (via Extern) — *Remote*

- Applied NLP and sentiment analysis on 1,000+ reviews using Python (pandas, NumPy) to surface key signals driving consumer preference – similar to feature extraction for ranking models.
- Identified 3 key sentiment drivers and delivered data-driven recommendations to product stakeholders.

Projects

Stocker – Stock Prediction & Ranking System | *Python, NLP, scikit-learn, Machine Learning*

- Trained a stock movement prediction model combining historical price data with NLP-based news sentiment features; applied feature engineering and walk-forward validation to achieve 62–65% directional accuracy.
- Reduced RMSE by 8% over baseline via cross-validation and iterative feature selection – core ML training and optimization workflow.

OPS Connect – Full-Stack Healthcare Portal | *React, Node.js, Flask, REST API*

- Built end-to-end application with authentication, dashboards, and RESTful API integration for 50+ patient records; delivered from wireframes to deployment in 3 months.

Technical Skills

ML & Modeling: Model training, feature engineering, cross-validation, embeddings, vector search, similarity/relevance scoring, RAG pipelines, LLM integration, NL-to-SQL, NLP, sentiment analysis, text classification
Frameworks & Libraries: scikit-learn, TensorFlow, PyTorch, pandas, NumPy, OpenAI API, Anthropic API
Languages: Python, SQL, JavaScript, TypeScript, Java
Cloud & Infra: AWS (EC2, S3, Lambda), Google Cloud, Docker, Git, Agile/Scrum
Backend & APIs: Flask, FastAPI, Node.js, REST APIs, Firebase
Data Viz: Matplotlib, Seaborn, Power BI
Compliance: CCPA, FERPA